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T12 – AI&DS

SUB : ASD&D Lab

EXPERIMENT – 4

AIM : To set up a Continuous Integration Pipeline using Jenkins.

LO Mapped : LO3

LO3 : Build and manage automated CI/CD pipelines for application development, testing, packaging ,and deployment.

Theory :

Continuous Integration is a development practice that promotes collaboration, early issue detection, and automated testing to enhance the quality and efficiency of software development.

Jenkins is an open-source automation server that facilitates continuous integration and continuous delivery (CI/CD) of software projects. It helps developers automate various stages of the software development lifecycle, from building and testing to deployment and monitoring. Jenkins supports a wide range of plugins and integrations, making it a versatile tool for automating various tasks in software development.

What is a Pipeline in Jenkins?

A Jenkins Pipeline is a set of automated processes that help developers and DevOps teams build, test, and deploy code in a Continuous Integration/Continuous Delivery (CI/CD) environment.

In simple terms, it is a workflow that defines how software moves from code to production in an automated and repeatable manner.

Key Benefits of Jenkins Pipeline

- Automates CI/CD processes.
- Provides better visibility into the build process.
- Offers version control of build logic (Pipeline as Code).
- Supports complex workflows such as parallel execution, conditional logic, and automated deployment.

Types of Jenkins Pipelines

A Jenkins Pipeline automates the complete CI/CD workflow using a **Jenkinsfile**.

A. Declarative Pipeline

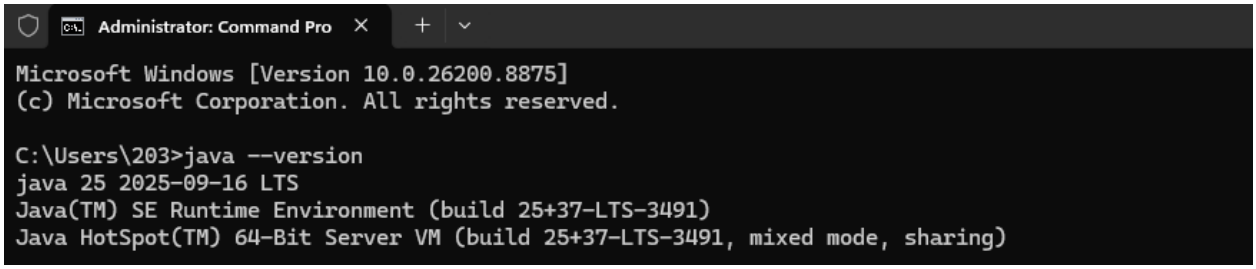
- Written inside the pipeline {} block.
- Simple, structured, and easy to read.
- Provides built-in syntax validation.
- Uses sections like agent, stages, stage, and steps.
- Best suited for standard CI/CD workflows.

B. Scripted Pipeline

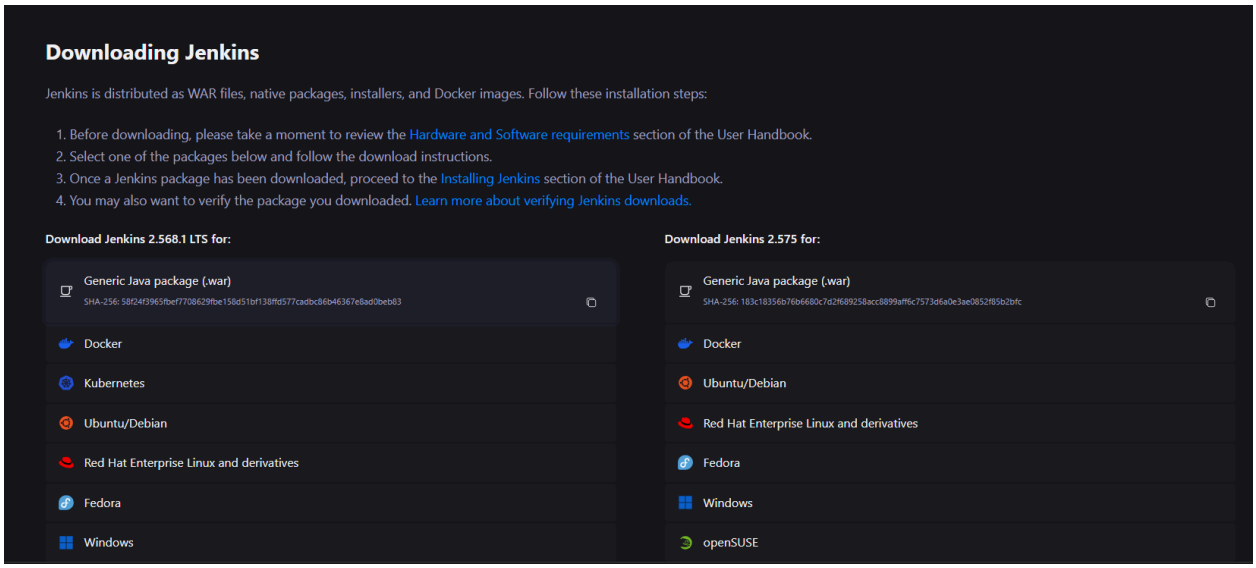
- Written inside the node {} block.
- Uses the Groovy scripting language.
- Highly flexible and customizable.
- Suitable for complex workflows with conditions and loops.
- More difficult to learn than Declarative Pipeline.

Jenkins Installation and setting up a continuous integration in pipeline :

Java Installation: Ensure Java Development Kit (JDK) is installed on your system.



Download Jenkins: Get the latest Jenkins WAR file from the official website or package manager.



Install Jenkins: Run the installer (or use the WAR file) and complete the installation process.

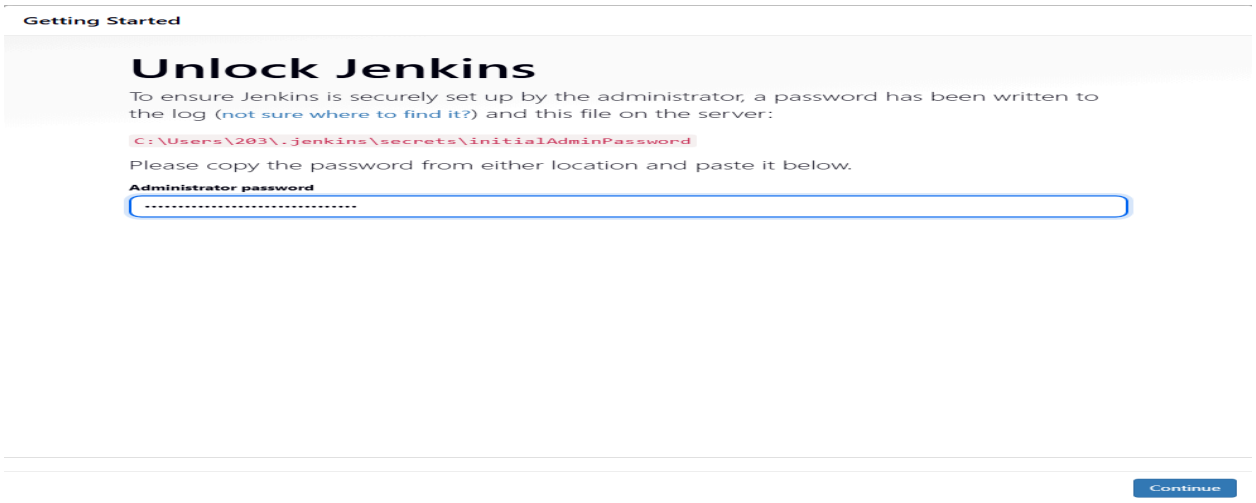
Start Jenkins: Launch the Jenkins service and open http://localhost:8080 in a web browser.

```
C:\Users\203\Desktop>java -jar C:\Users\203\Desktop\jenkins.war --httpPort=8081
Running from: C:\Users\203\Desktop\jenkins.war
webroot: C:\Users\203\.jenkins\war
2026-08-04 05:07:41.587+0000 [id=3]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-08-04 05:07:41.623+0000 [id=3]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-08-04 05:07:41.662+0000 [id=3]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.8; built: 2026-04-01T01:09
2026-08-04 05:07:42.194+0000 [id=3]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did
2026-08-04 05:07:42.230+0000 [id=3]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by com.thoughtworks.xstream.converters.reflection.SunUnsafeReflect
WARNING: Please consider reporting this to the maintainers of class com.thoughtworks.xstream.converters.reflection.SunUnsafeRe
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
2026-08-04 05:07:42.593+0000 [id=3]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\203\.je
2026-08-04 05:07:42.679+0000 [id=3]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreConte
ABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@11ee02f8{STARTED}}
2026-08-04 05:07:42.710+0000 [id=3]     INFO    o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@1c65121{HT
2026-08-04 05:07:42.710+0000 [id=3]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@57dc9128{STARTING
2026-08-04 05:07:42.711+0000 [id=67]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disab
2026-08-04 05:07:42.812+0000 [id=60]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.568.1
2026-08-04 05:07:42.889+0000 [id=76]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-04 05:07:42.898+0000 [id=96]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-04 05:07:43.461+0000 [id=105]   INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-04 05:07:43.466+0000 [id=107]   INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-04 05:07:43.467+0000 [id=88]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-04 05:07:43.566+0000 [id=84]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-04 05:07:43.573+0000 [id=92]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-04 05:07:43.574+0000 [id=96]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-04 05:07:43.575+0000 [id=96]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-08-04 05:07:43.585+0000 [id=127]   INFO    hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-08-04 05:07:43.767+0000 [id=112]   INFO    jenkins.install.SetupWizard#init:
[CRLF]>
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]>
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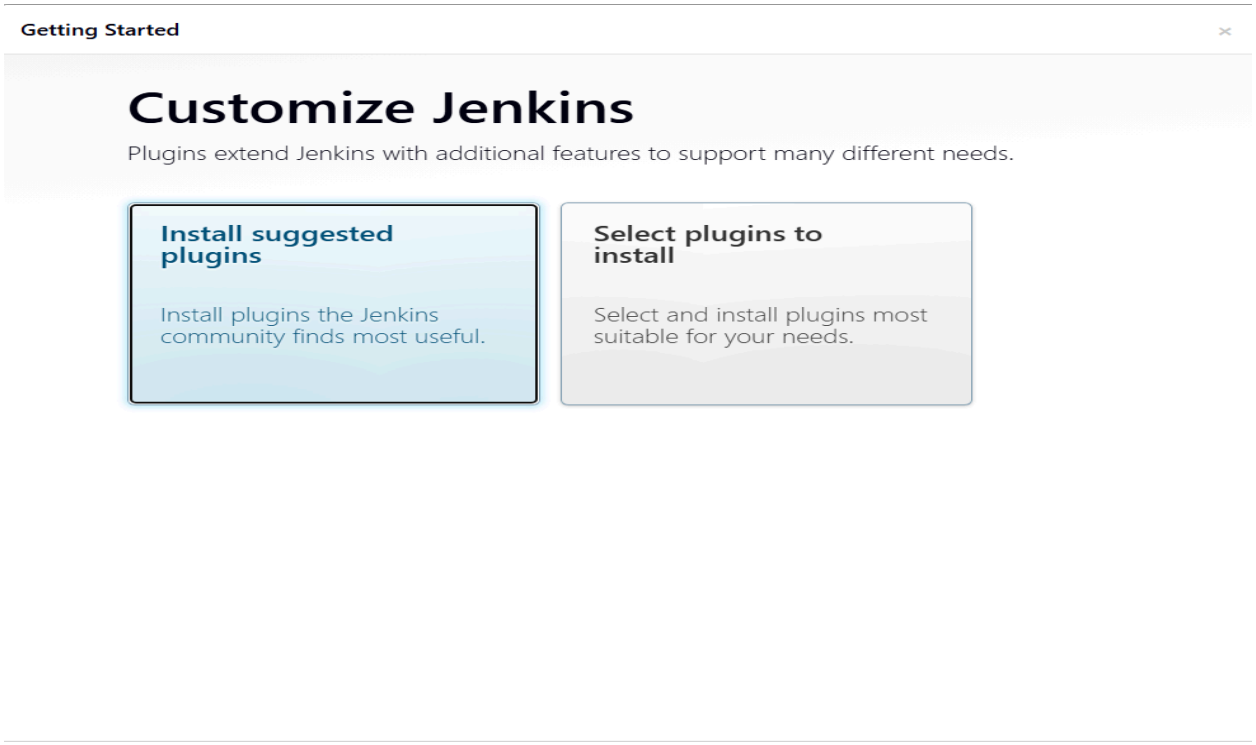
```
[CRLF]>
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]>
[CRLF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[CRLF]> Please use the following password to proceed to installation:
[CRLF]>
[CRLF]> 7ecb430f89034d5b9e8d8ce22246fce4
[CRLF]>
[CRLF]> This may also be found at: C:\Users\203\.jenkins\secrets\initialAdminPassword
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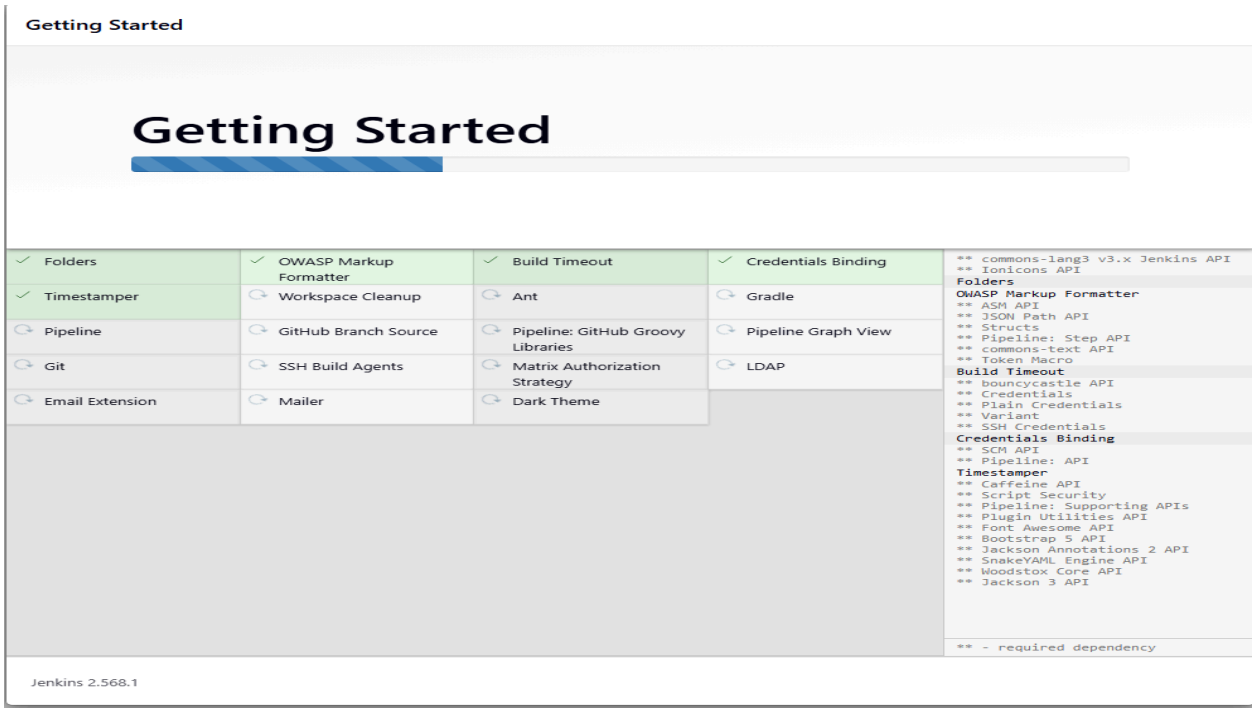
2026-08-04 05:07:47.376+0000 [id=89]    INFO    jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-08-04 05:07:47.394+0000 [id=60]    INFO    hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2026-08-04 05:07:48.830+0000 [id=127]   INFO    h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
```

Unlock Jenkins: Enter the initial administrator password.

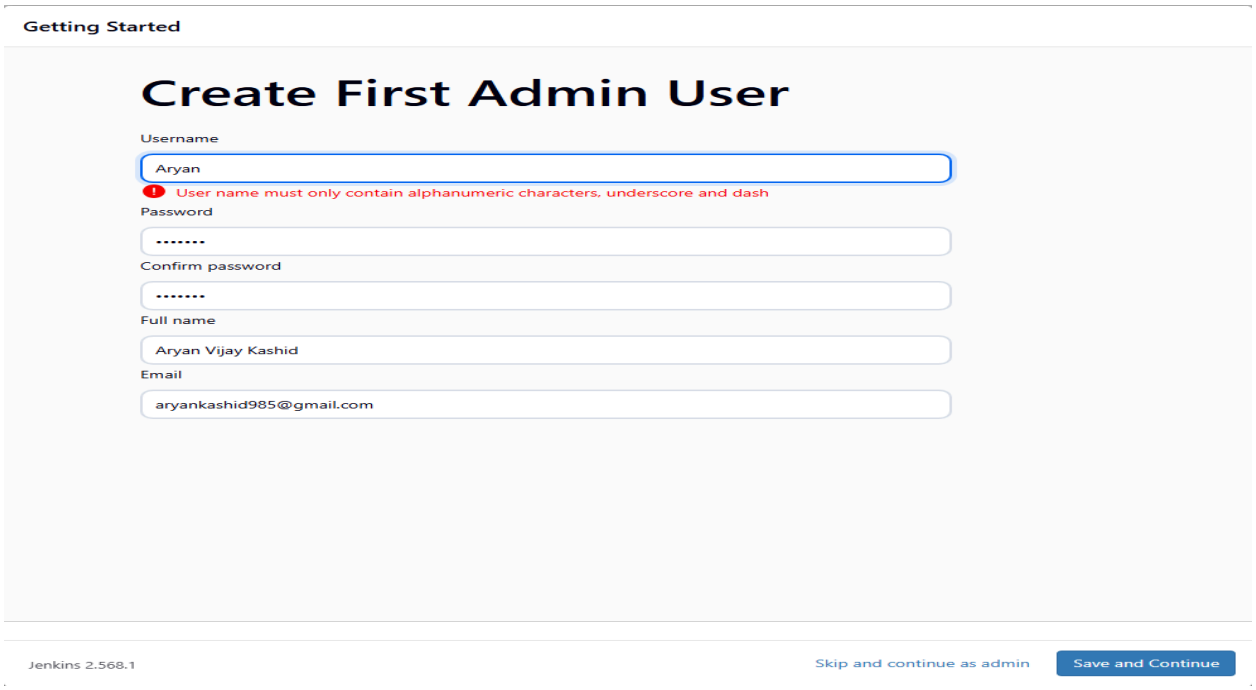


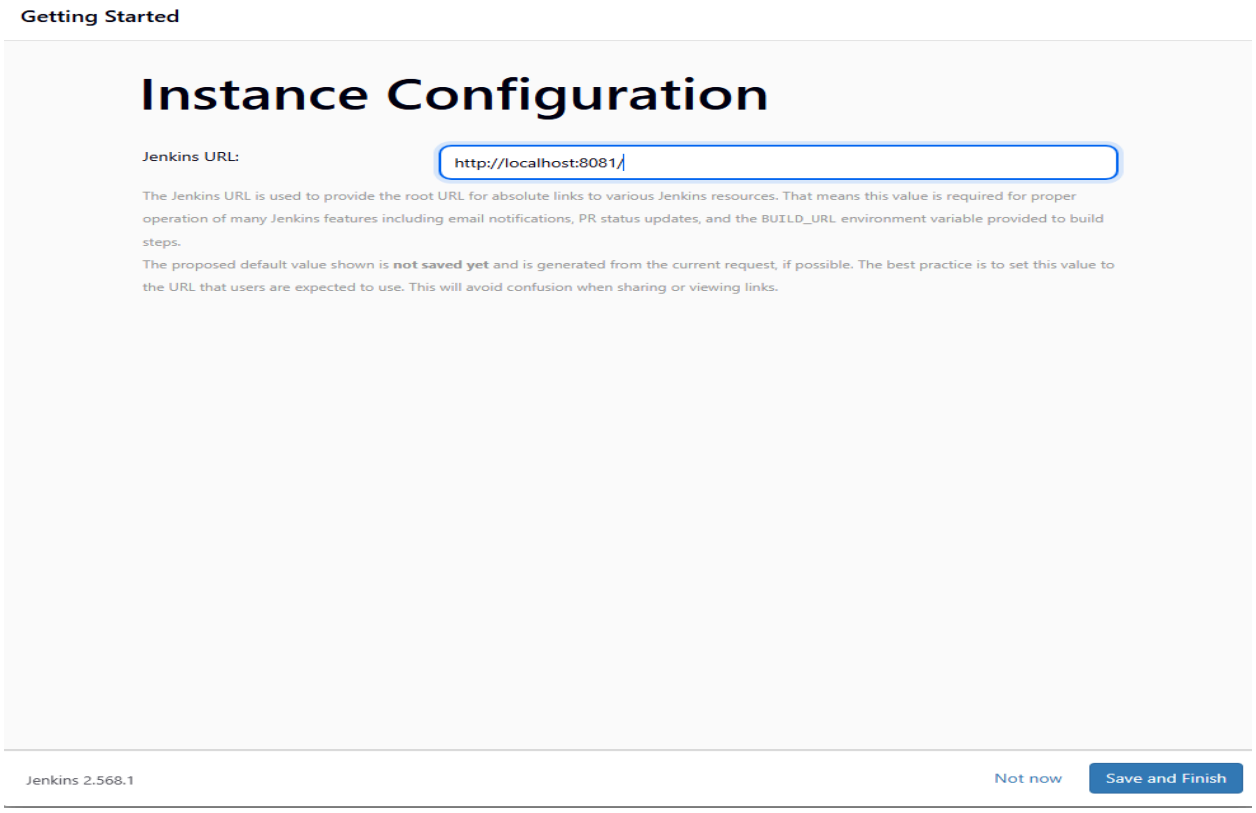
Install Suggested Plugins: Choose the suggested plugins.

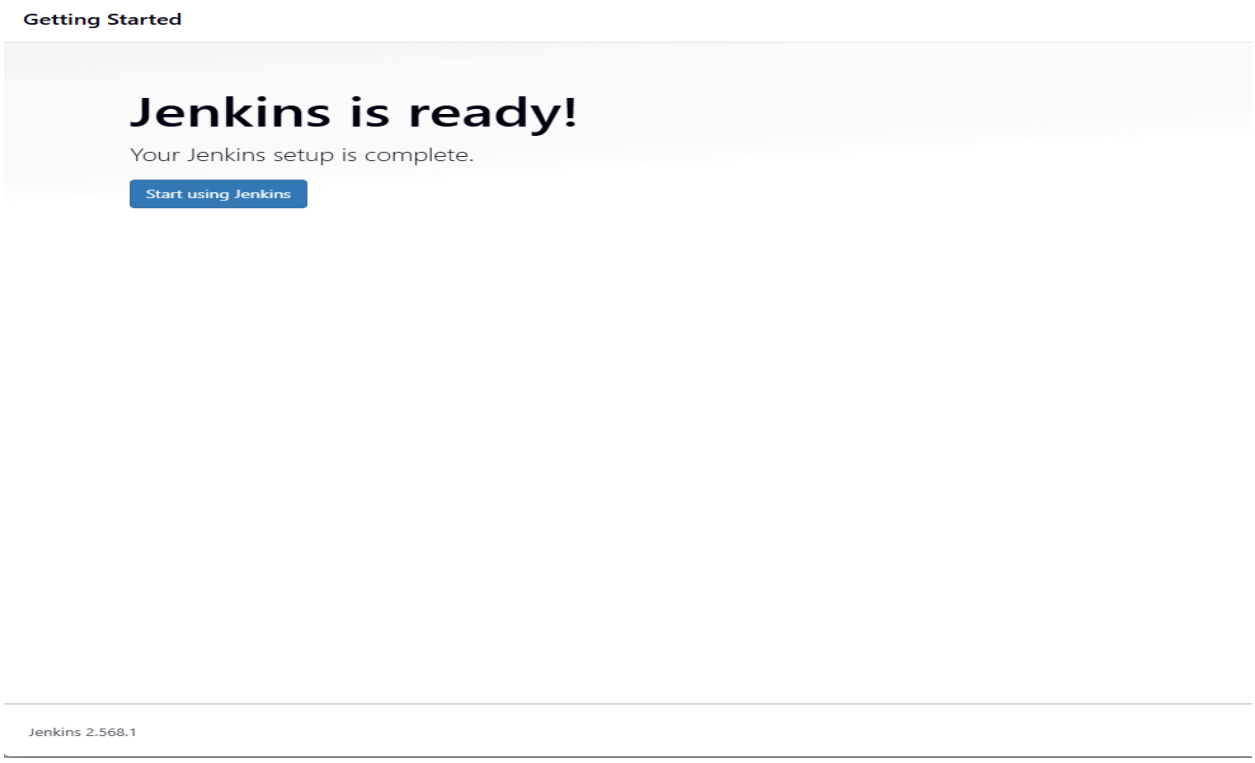




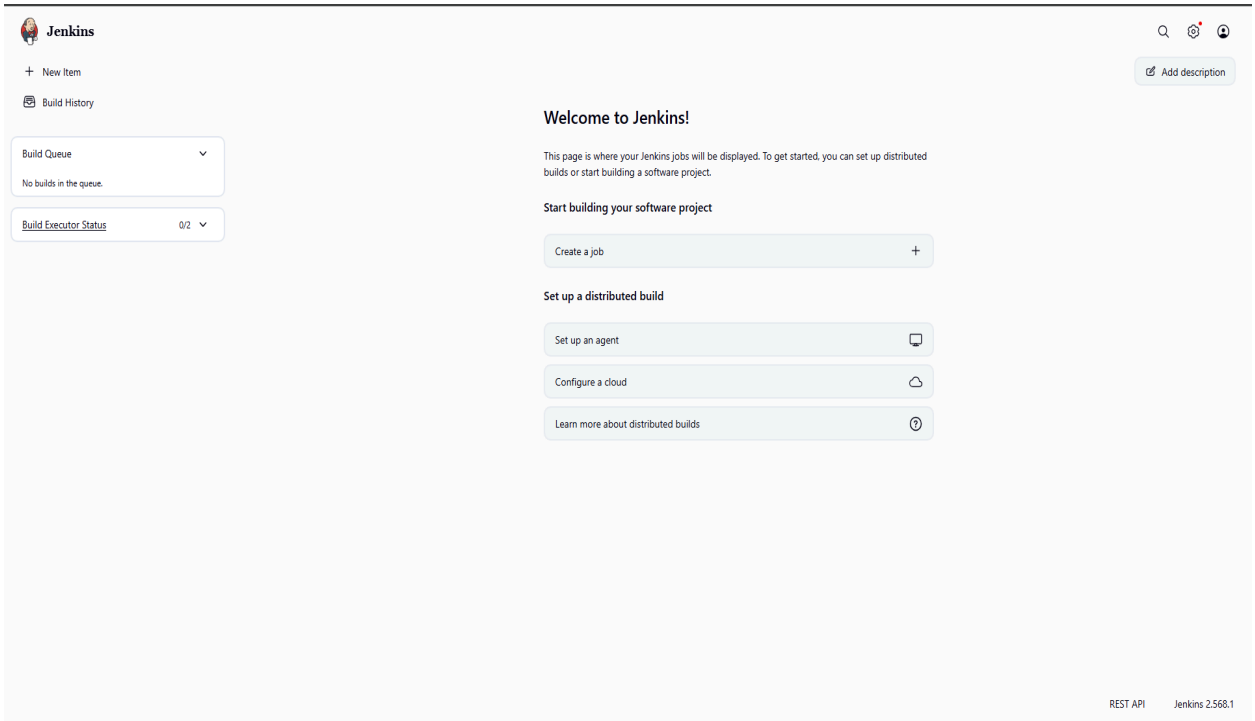
Create Admin User: Create the first administrator account.

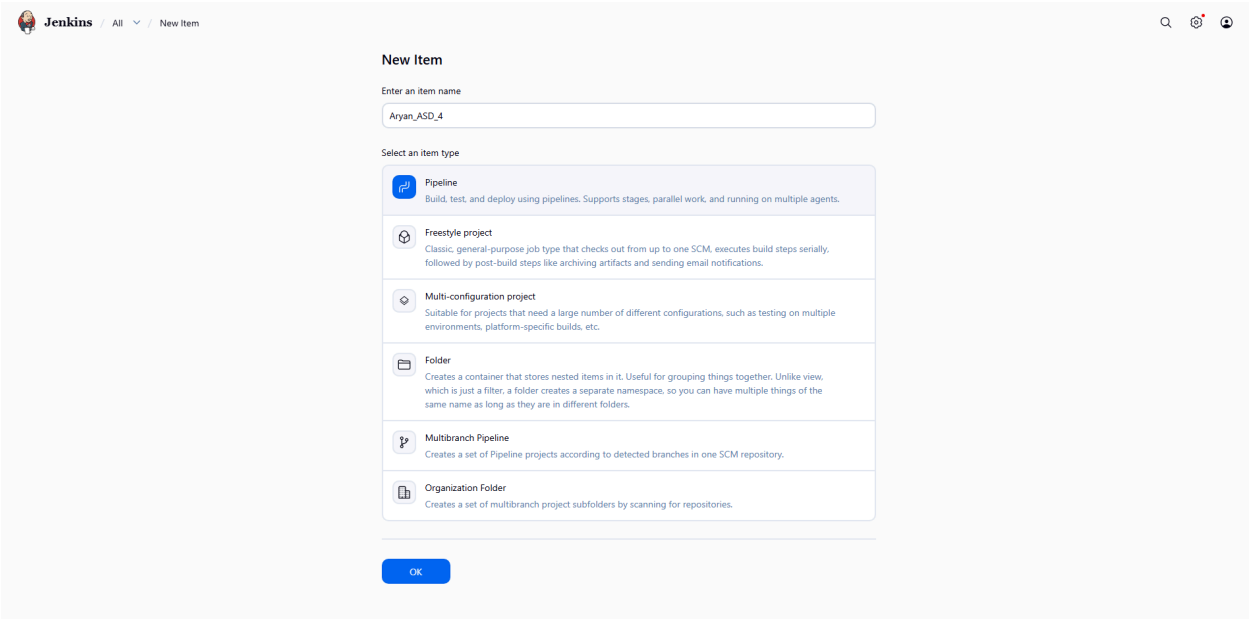




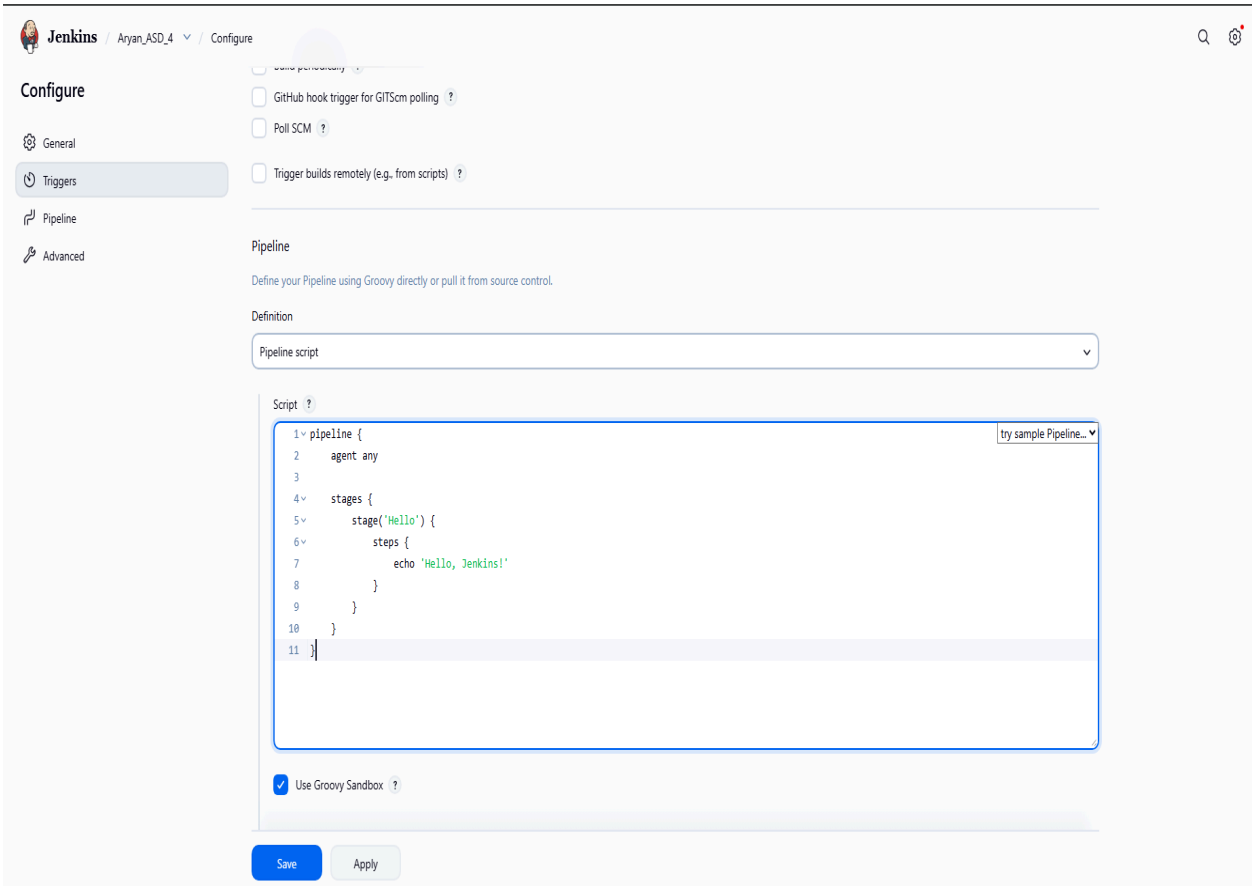


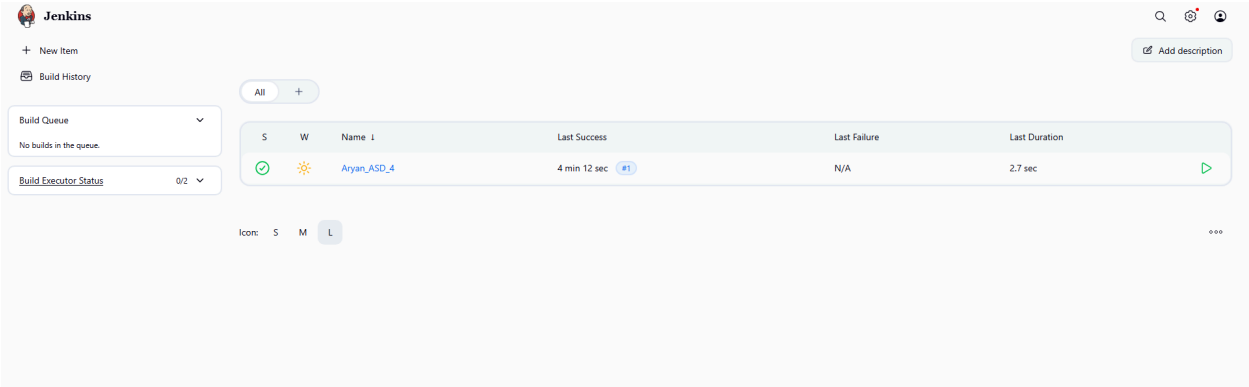
Create a New Job: Create a Freestyle project.



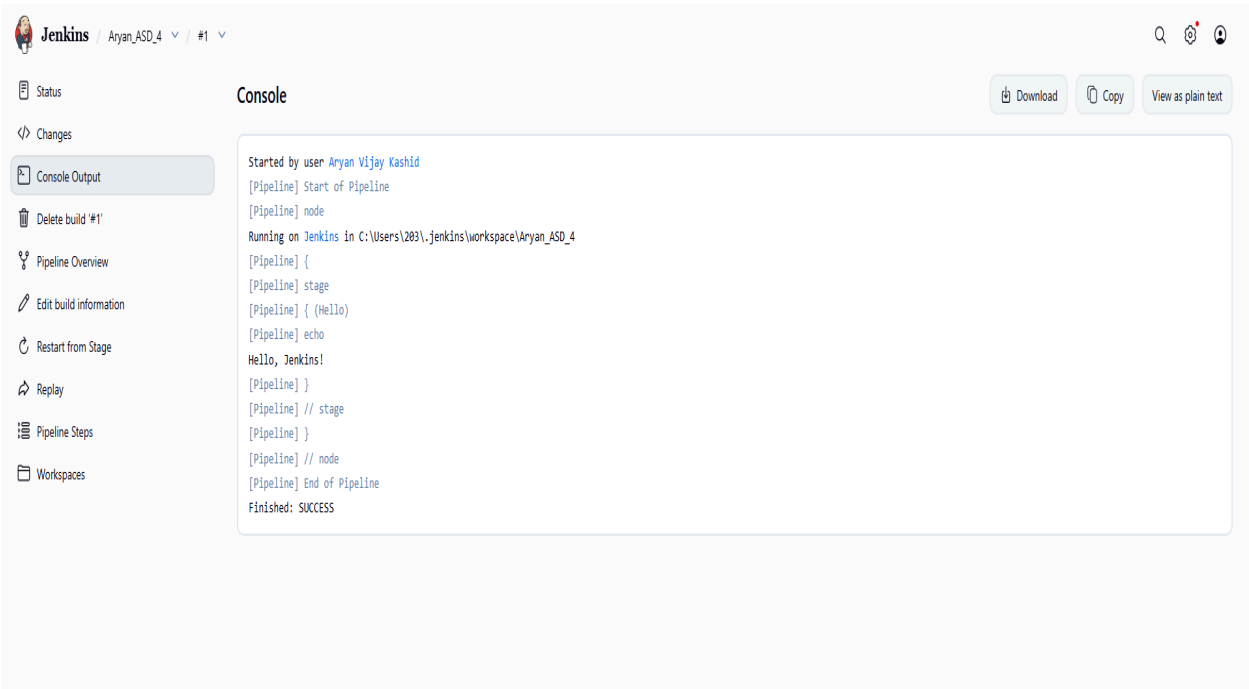


Configure Build Steps: Add build commands/scripts.





Build the Project: Click 'Build Now' and verify output.



Conclusion :

In this experiment, the Jenkins Pipeline was successfully implemented to automate the Continuous Integration (CI) process. It streamlined the build and execution workflow, reduced manual effort, and ensured faster, more reliable software development through automation.